1. 2012Q35
Which of the following belongs to humoral immune response?
A. blood clotting
B. production of antibiotics
C. phagocytosis of pathogens
D. production of memory T cells

2. 2012Q36
After injury, the wound usually becomes swollen due to:
A. accumulation of bacteria at the wound.
B. accumulation of tissue fluid at the wound.
C. increased phagocytosis at the wound.
D. increased blood flow to the capillaries around the wound.

3. 2014Q2
Which of the following events ***does not*** involve the functioning of membrane proteins?
A. Transmission of nerve impulses across a synapse
B. Absorption of glucose in the small intestine
C. Transport of oxygen by haemoglobin
D. Recognition of pathogens

4. 2014Q32
Infants can obtain antibodies from breast feeding. Which of the following combinations correctly describes this type of immunity in infants?

	Type of immunity	***Explanation***
A.	active	the antibodies are produced from white blood cells
B.	active	the antibodies attack pathogens bearing foreign antigens
C.	passive	the antibodies do not trigger the production of memory cells
D.	passive	the antibodies work only when there is re-entry of the same pathogen

5. 2015Q32
Questions 32 refer to the table below, which shows the results of blood tests for the presence of antigens and antibodies of hepatitis B in four individuals:

	Individual 1	Individual 2	Individual 3	Individual 4
Antigens of hepatitis B	Negative	Positive	Negative	Positive
Antibodies of hepatitis B	Negative	Negative	Positive	Positive

Which individual(s) would you recommend for vaccination against hepatitis B?
A. 1 only
B. 4 only
C. 1 and 2 only
D. 1 and 3 only

6. 2015Q36
Which of the following components of blood are involved in forming a blood clot?
(1) Blood platelets
(2) Red blood cells
(3) White blood cells
A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

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7. 2018Q10
Which of the following statements best explains why vaccination against flu is administered annually?
A. The flu virus is constantly mutating.
B. The antibodies against flu virus only last for one year.
C. The flu vaccine is not very effective because it is made from a weakened virus.
D. When memory cells encounter the flu vaccine again, a secondary response is triggered.

8. 2019Q36
Antibodies are produced by
A. memory B cells.
B. memory T cells.
C. specialized B cells.
D. specialized T cells.

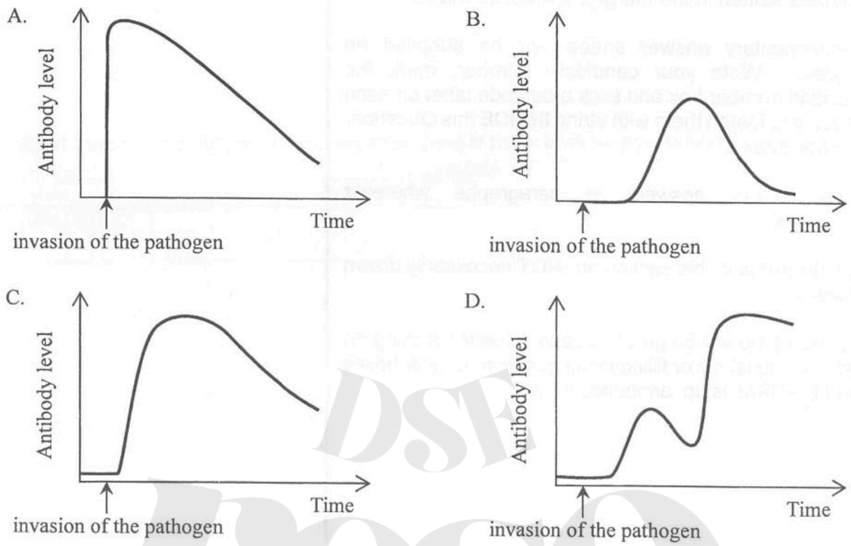
9. 2020Q30
Which of the following provides immunity to the human body?
A. phagocytosis
B. inflammation
C. memory cell
D. formation of blood clots

10. 2021Q24
Which of the following types of cells can be found in the tissue fluid?
(1) B cell
(2) T cell
(3) phagocyte
A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

11. 2021Q25
Which of the following is ***not*** the first line of defence against the invasion of pathogens in humans?
A. saliva
B. sweat
C. mucus
D. lymph

12. 2023Q36

A person is successfully vaccinated against infectious disease P. Which of the following graphs correctly shows the change in the antibody level of this person when he is invaded by the pathogen of disease P?



13. 2024Q36

Which of the following are examples of non-specific defence in humans?

(1) tear glands
(2) phagocytes
(3) epithelial tissue

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

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LQ

1. 2012Q1

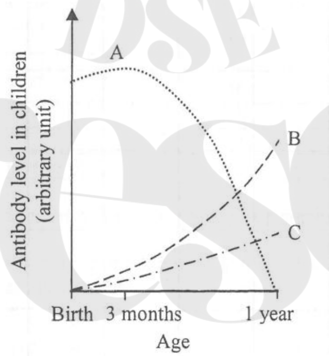
For each type of the blood cells listed in column 1, select from column 2 ***one*** phrase that correctly describes its function. Put the appropriate letter in the space provided. (3 marks)

Column 1		Column 2	
Lymphocytes	______	A.	Involved in blood clotting
Blood platelets	______	B.	Involved in oxygen transport
Red blood cells	______	C.	Involved in antibody production
		D.	Involved in phagocytosis
		E.	Involved in transporting hormones

2. 2013Q9

a) The following graph shows the change in levels of antibodies in children's bodies:

(A) antibodies from mother
(B) children's own antibodies with vaccination
(C) children's own antibodies without vaccination



i) State the types of immunity resulting from A, B and C. (3 marks)

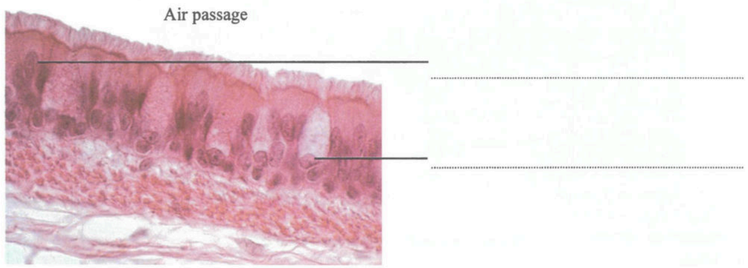
A: B: C:

ii) Suggest ***two*** possible ways that newborns can acquire antibodies from their mother. (2 marks)

iii) Explain why children who have been vaccinated against diseases are better protected than those who have not. (4 marks)

3. 2014Q2b

The photomicrograph below shows a section of the inner wall of the human trachea:

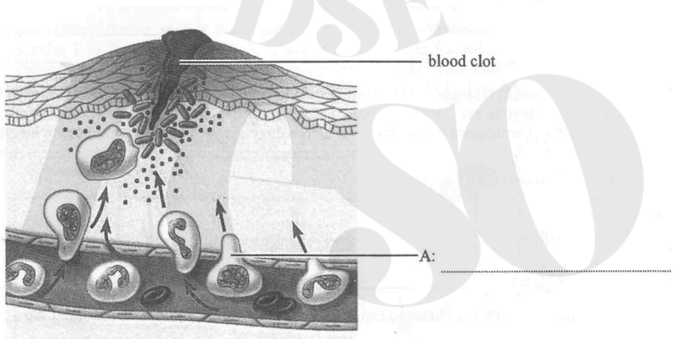


(x 400)

b) With reference to the features of the inner wall shown in the photomicrograph, describe how the inner wall of the trachea can protect our body against bacterial invasion. (3 marks)

4. 2016Q7

The diagram below shows a site of injured skin exhibiting an inflammatory response:



a) Label the type of white blood cell represented by cell A in the above diagram. (1 mark)

b) Explain why the tissue exhibiting the inflammatory response usually shows symptoms such as redness, swelling and pain. (3 marks)

c) Cell A will present the antigens of the invading pathogens to the lymphocytes. Describe what will happen subsequently. (3 marks)

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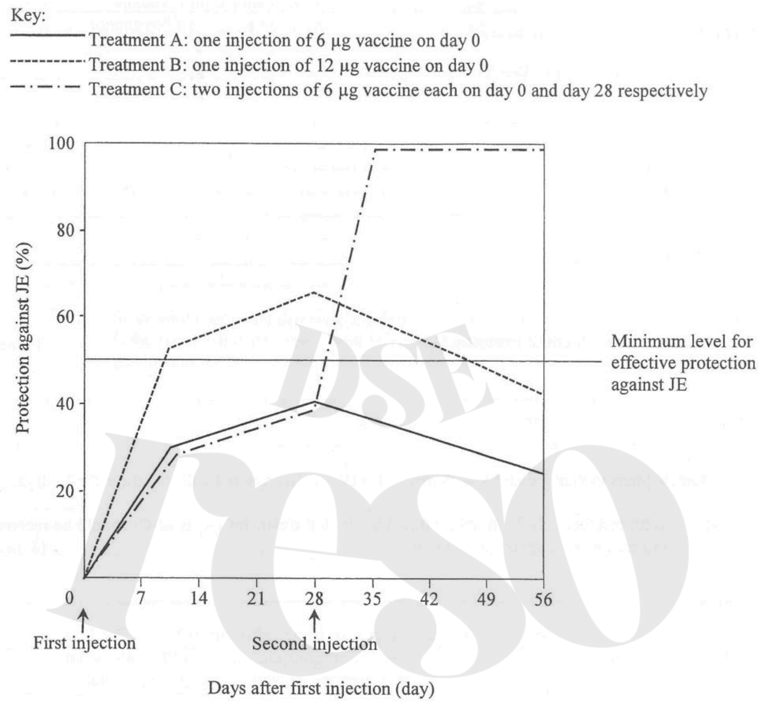
5. 2017Q9b, c, d

Japanese encephalitis (JE) is an inflammation of the brain caused by a viral infection. Scientists have developed a vaccine against the JE virus. In a study of the effectiveness of the vaccine, three groups of healthy people received different vaccination treatments and the level of protection against JE was monitored over a period of time. The results are shown in the graph below:



b) For treatment C, explain why there is a sharp rise in protection against JE from day 28 to day 35. (4 marks)

c) Give ***one*** more benefit of treatment C. (1 mark)

d) Mathew plans to visit a country with many JE cases in 10 days and will stay there for 15 days.

i) With reference to the graph, which vaccination treatment (A, B or C) should he receive at this moment? Explain your answer. (2 marks)

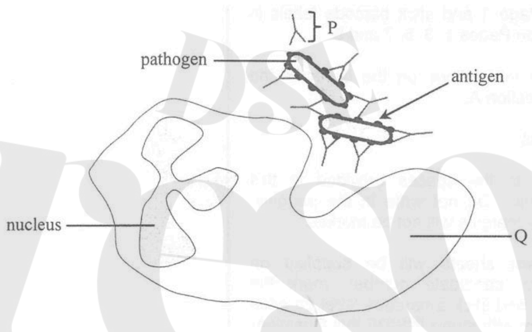
ii) As a responsible citizen, Matthew will continue to use insect repellent as a precaution for two weeks after he is back from that country. Suggest a rationale for this precaution. (1 mark)

6. 2019Q1

a) Physical and chemical barriers are the first line of defence in the human body. Select from Column II ***all*** correct example(s) that belong(s) to the two types of barriers in Column I and put the letter(s) in the spaces provided. (2 marks)

	Column 1		***Column 2***	
(i)	physical barrier	________	A.	skin
			B.	tear
(ii)	chemical barrier	________	C.	antibody
			D.	blood clot
			E.	gastric juice

b) The diagram below shows the process of phagocytosis. Q is a phagocyte while P is a protein molecule produced by a type of lymphocyte.



Describe the function of P in phagocytosis. (3 marks)

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7. 2020Q8

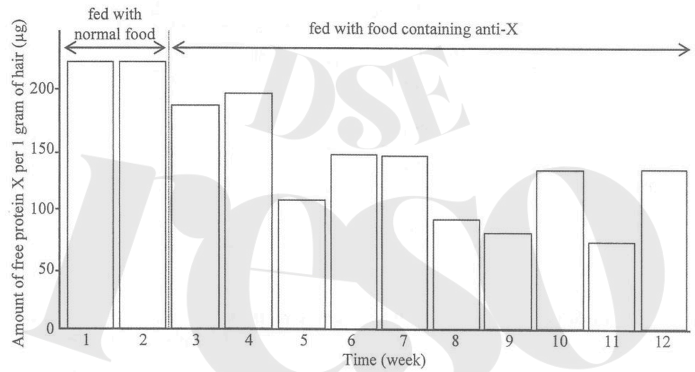
Some people suffer from sneezing and coughing when there are cats nearby. These unwanted immune responses, known as allergies, are caused by a protein X secreted by cats' salivary and sebaceous glands. When cats lick their body surface, this protein X is spread to their hair. Protein X can accumulate in their living environment over time.

a) Recent research shows that the amount of free protein X on cats' hair can be reduced by adding antibodies against protein X (anti-X) to cat food.

i) Why can anti-X reduce the amount of free protein X on cats' hair? (1 mark)

ii) There are no proteases in the saliva of cats. Explain why this is important for the success of this method of reducing free protein X. (1 mark)

b) In the research, a group of domestic cats were fed with normal cat food for 2 weeks, followed by cat food containing anti-X for 10 weeks. The amount of free protein X found on their hair during the research is shown in the bar chart below:



i) Research team member A thought anti-X was effective in reducing the amount of free protein X on cats' hair while team member B thought anti-X was not effective. Based on the bar chart, give ***one*** reason to support team member A and ***another*** reason to support team member B. (2 marks)

ii) How would you modify the research to confirm whether anti-X in cat food is effective in reducing the amount of free protein X on cats' hair? (2 marks)

c) Based on the information in the introductory paragraph of this question, suggest ***two*** limitations of this approach to reducing allergies caused by cats. (2 marks)

8. 2021Q6a, b

Pathogen X is a pathogen that infects humans. Research has discovered an antigen Y present on the surface of pathogen X. Using recombinant DNA technology, antigen Y can be produced and serves as a vaccine to induce immunity against pathogen X.

a) Explain how the injection of antigen Y can induce immunity against pathogen X. (4 marks)

b) Other than the use of recombinant DNA technology, suggest another way to produce a vaccine. (1 mark)

9. 2023Q4c

Dengue fever is an infection caused by the dengue viruses (DENV). It is an endemic illness in many countries in tropical and sub-tropical regions. DENV encompasses four different subtypes. Each subtype can lead to dengue fever.

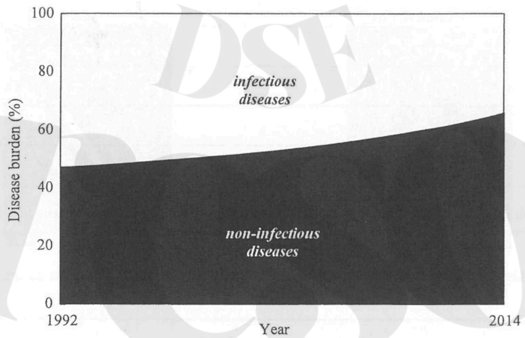
c) Patients infected with a particular subtype of DENV for the first time can recover on their own after about a week without any treatment.

i) Give ***three*** types of white blood cells that aid the recovery and describe each of their actions. (3 marks)

ii) Explain why people who have recovered from infection with a particular subtype of DENV can still be infected with other subtypes of DENV in the future. (2 marks)

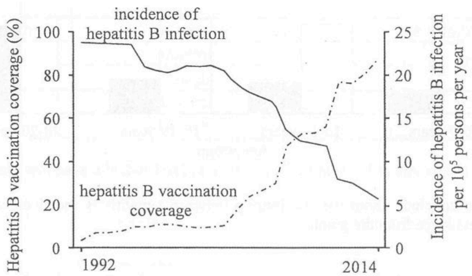
10. 2024Q8

Disease burden is a measure of population health that aims to quantify the potential loss of lifespan and health outcomes due to illness as compared to the ideal of living to a ripe old age and in good health. The graph below shows the percentage share of disease burden caused by infectious diseases and non-infectious diseases in Country X from 1992 to 2014:



a) Describe the change in percentage shares of the disease burden of Country X from 1992 to 2014. (1 mark)

b) The graph below shows the impact of hepatitis B vaccination on the incidence of hepatitis B infection in Country X from 1992 to 2014:



With reference to the principle of vaccination, explain the relationship shown in the above graph. (4 marks)

c) With reference to the information from (a) and (b), suggest the role of vaccination in the change of disease burden in Country X. (1 mark)

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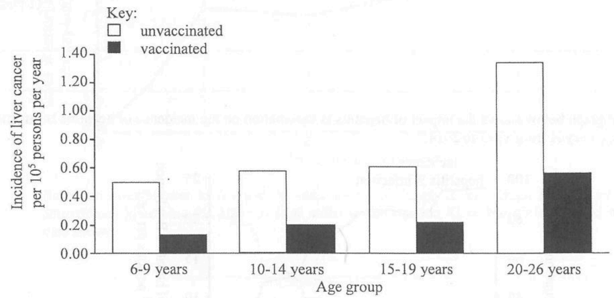
d) The graph below shows the incidence of liver cancer among different age groups who have or have not been vaccinated against hepatitis B in Country X:



What can you conclude about the relationship between hepatitis B and liver cancer? Support your answer with evidence from the graph. (2 marks)

Ans

2012Q35: B (71%)	2012Q36: B (70%)	2014Q2: C (32%)	2014Q32: C (61%)
2015Q32: A (38%)	2015Q36: D (34%)	2018Q10: A (78%)	2019Q36: C (66%)
2020Q30: C (65%)	2021Q24: D (66%)	2021Q25: D (65%)	2023Q36: C (70%)
2024Q36: D (43%)			

2012Q1

1 C (1)
A (1)
B (1)

3 marks

2013Q9

9 (a) (i) A: passive immunity (1)
B: active immunity (1)
C: active immunity (1) (3)

(ii)
- some antibodies in the maternal blood pass through the placenta and enter into the foetal blood (1)
- some maternal antibodies in the mother's milk pass to the newborn via breast feeding (1) (2)

(b)
- vaccine contains antigens (1)
- which stimulate the immune system to produce memory cells for that particular antigens (1) (4)
- on the second exposure to the same antigen (1)
- these memory cells are capable of producing a large amount of antibodies (1)

therefore, child with vaccination has a better protection

9 marks

2014Q2b

2 (b)
- mucus secreting cells secrete mucus to trap germs / pathogens / bacteria / microbes from the incoming air (1)
- cilia sweep the trapped germs away to the throat for swallowing or coughing (1) (3)
- closely packed epithelial cells prevent the entry of bacteria / form a physical barrier (1)

2016Q7

5 (a) phagocyte (1) (1)

(b)
- arterioles of the tissue with inflammatory response dilate, increasing blood flow to the tissue and makes it red (1)
- permeability of capillary wall increases, increasing the formation of tissue fluid and its accumulation, and leads to swelling (1) (3)
- more tissue fluid presses against nerve endings, stimulating the pain receptors and gives the pain sensation (1)

(c)
- activity of B-lymphocytes will lead to the production of antibodies (1) against the specific pathogens
- activity of T-lymphocytes will lead to the destruction of infected cells (1) (3)
- memory cells will be formed for future immunity / quicker response in the second attack (1)

7 marks

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2017Q9b, c, d

9 (b)
- certain lymphocytes differentiated into memory cells when they encountered the antigens of the vaccine in the first injection (1)
- in the second injection, these memory cells encountered the same antigens again (1)
- these memory cells differentiated into specific B-lymphocytes / specific T-lymphocytes (1) (4)
- resulting in production of a large amount of antibodies / killer T-cells within a shorter time (1)

leading to the sharp rise in the protection against the infection

(c)
- the protective effect does not wear off / remains high (1) from day 35 to 56 (1)

(d) (i)
- vaccination treatment B (1)
- as it offers protection over the minimum level of effective protection from day 10 to day 47 which fully covers Mathew's trip (1) (1+1)

(ii)
- even if he has contracted JE during the trip, the precaution can help prevent / reduce the risk of transmission of the virus (1) to other people as the insect repellant prevents mosquito bite. (1)

2019Q1

5 (a) (i) A, D (1)
(ii) B, E (1) (2)

(b)
- P represents antibodies which attach to the antigens on the surface of the pathogens (1)
- as a result, P binds several pathogens together as a big mass / clump (1) (3)
- to enhance / facilitate the phagocytosis by Q (1)

5 marks

2020Q8

8 (a) (i) anti-X can bind to protein X to form clumps / agglutination (1) — (1)

(ii) anti-X in the food debris left in the mouth cavity of the cats will not be digested (1), as a result, it can reduce the amount of free protein X in their saliva — (1)

(b) (i)
- A: as the data show a decreasing trend/drop in the amount of protein X found (1)
- B: as the data show fluctuations of comparable levels (1) from week 5 to week 10

(2)

(ii)
- conduct the research for a longer time (1)
- to check if the decreasing trend will continue / if the fluctuations will continue (1)

OR
- conduct the research with a higher dose of anti-X in cat food (1)
- to check if there will be a greater decrease in the amount of free protein X on cats' hair (1)

OR
- conduct statistical analysis of the data (1)
- to check if the difference shown is significant or not (1)

OR
- conduct the research with a greater sample size (1)
- to check if similar trends / patterns in the change of the amount of protein X can be observed (1)

OR
- set up a control with normal food / boiled anti-X in cat food (1)
- to check if there is still a decrease in the amount of free protein X on hair / to compare the result with the experimental group (1)

OR
- conduct the research with groups of different types of cats (1)
- to check if there will be a greater decrease in the amount of free protein X on cats' hair for certain type of cats (1)

(1+1)

(c)
- can only reduce the amount of protein X in saliva but not that produced by sebaceous glands (1)
- protein X can accumulate in the environment, its amount can finally reach a critical level for eliciting unwanted immune responses (1)

(2)

8 marks

2021Q6a, b

6 (a)

Concept for mark award
- recognition of antigen Y as a foreign antigen (1)
- production of memory cells for antigen Y (1)
- 2nd encounter of antigen Y on pathogen X (1)
- elicitation of secondary immune response (point out the characteristics of the secondary immune response) (1)

(4)

e.g.
- after injection of antigen Y into the human body, B-lymphocytes will recognise antigen Y as a foreign antigen (1)
- the B-lymphocytes will produce memory cells for antigen Y (1)
- when pathogen X bearing antigen Y invades the body in the future, these memory cells can recognise the same antigen (1)
- and elicit a secondary immune response that produces a large number of antibodies in a short time (1) to eliminate the invading pathogen X

(b)
- vaccination derived from the weakened pathogen (1) — (1)

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2023Q4c

4 (c) (i)
- B lymphocytes / plasma cells produce antibodies to neutralise the dengue viral particles / aggregate the viral particles, preventing them from entering the body cells (1)
- phagocytes engulf the aggregated viral particles (1)
- T lymphocytes recognise and kill the body cells which have been infected by the virus (1)

(3)

(ii)
- the different subtypes of DENV may have antigens different from one another (1)
- as a result, the memory cells produced after the infection of one subtype may not be able to recognise antigens of other subtypes (1)

(2)

2024Q8

	Concept for mark award	Example	Marks
8(a)	description of the trends shown	the disease burden caused by infectious diseases decreases while that caused by non-infectious diseases increases	(1)
8(b)	correct causal relationship	incidence of hepatitis B infections decreases with the increase in vaccination coverage	(1)
	principle of vaccination ● 1st exposure to antigen ● production of memory cell ● re-exposure to the same antigen leading to secondary response	vaccination allows pre-exposure of the immune system / B cells / T cells / lymphocytes to antigens of the hepatitis B virus	(1)
		this produces memory cells for the antigens of hepatitis B virus	(1)
		when the memory cells encounter the antigens of the real hepatitis B virus, the memory cells elicit a secondary response / a stronger or faster immune response to eradicate the hepatitis B virus before it can invade the cells	(1)
8(c)	relate the decrease in disease burden to decrease in incidence	vaccination programmes can reduce the incidence of infectious diseases / reduce the number of susceptible hosts to break the chain of infection, thereby decreasing the disease burden caused by infectious diseases in country X	(1)
8(d)	correct description of data with reference to all age groups	the incidence of liver cancer is higher in the unvaccinated group than the vaccinated group in all age groups	(1)
	relate the data trend to concept of risk factor	hepatitis B is a risk factor for liver cancer / having hepatitis B increases the risk of getting liver cancer	(1)